

SUJIT PERAMANU

☎ 403-969-6430 ✉ sujit.peramanu@mail.utoronto.ca [in ca.linkedin.com/in/sujit-peramanu](https://ca.linkedin.com/in/sujit-peramanu) | sujitperamanu.com

Education

B.A.Sc - Engineering Science in Robotics + AI Eng. Minor + PEY Co-op

Sep 2021 – April 2026

University of Toronto, Toronto Ontario

Relevant Coursework

- | | | | |
|-------------------------|-------------------|--------------------------|--------------------|
| • Computer Vision | • Dynamics | • Mobile Robotics | • OOP |
| • Linear Control Theory | • Data Structures | • Reinforcement Learning | • Embedded Systems |

Technical Skills

Languages: C++, Python, C, MATLAB, TypeScript, JavaScript

Robotics & GNC: ROS2, Sensor Fusion, SLAM, State Estimation, Visual Odometry, Path Planning, PID/Linear Control, CV

Tools & Frameworks: Linux, Gazebo, TensorRT, OpenCV, Vue, React, Django

Work Experience and Research

Robotics Research Thesis at MEDCVR

September 2025 – Present

Institute of Biomedical Engineering at the University of Toronto

Toronto, Canada

- Designed and evaluated learning-based closed-loop control policies (Diffusion Policy, DP3, ACT) for constrained manipulation on a Stäubli TX2-60L 6-DoF robotic manipulator across absolute-joint, relative-joint, and Cartesian remote-center-of-motion control
- Built a real-time multimodal perception pipeline in ROS2 with POSIX shared-memory point-cloud ingestion at 20 Hz, deploying closed-loop inference at 10–15 Hz on physical hardware
- Engineered a two-stage remote-center-of-motion inverse-kinematics solver and applied VoxelNeXt point-cloud encoding for visuomotor state estimation from optical coherence (OCT)
- Achieved full task completion with robust subsurface-aware grasping using a policy conditioned solely on OCT point clouds; paper in preparation for journal submission

Robotics/Software Engineering Intern

November 2024 – August 2025

Mission Control Space Services

Ottawa, Canada

- Stood up a Java-built YAMCS mission server and developed a Python-Redis command-execution backend bridge; integrated both with Spacefarer™ for the Astrobotic CubeRover™ lunar demonstration mission
- Enhanced BEACON Bridge, flight software for live lunar rover operations, with command-action features and comprehensive unit testing
- Implemented an integrated mission activity-planner mode in Spacefarer™, the web-based platform for ground-segment space operations, using Vue.js and a Django authentication framework

Robotics Engineering Intern

July 2024 – October 2024

OHB System AG

Munich, Germany

- Developed KRL scripts using real-time RSI contexts to create force/torque-controlled tasks on a KUKA 6-axis robotic manipulator for OHB component-testing procedures
- Authored end-to-end documentation for the installation and programming of the KUKA ForTTran SG-500 force/torque sensor for internal company use

Software Engineer Intern

May 2022 – August 2022 | May 2023 – August 2023

Tektelic Communications

Calgary, Canada

- Designed and deployed gateway test automation scripts in Robot Framework, cutting end-to-end testing duration by 40%
- Built Python tooling to automate Ethernet BSP upgrade report generation; leveraged TestRail for developer-driven QA pipelines

Projects and Design Teams

Computer Vision Lead | Drone Robotics Capstone Design

January 2026 - April 2026

- Designed and deployed a stereo-vision pipeline on Jetson Nano using a custom-trained, TensorRT-optimized YOLOv8n model for real-time single-class detection at ~20 Hz within ± 2 cm pose error
- Fused IMX219 RGB detections with Intel RealSense T265 visual-inertial odometry and stereo depth (SGBM) via cross-camera extrinsic calibration to recover full 3D target pose; applied adaptive EMA smoothing and jump rejection for robustness under vibrational load
- Architected a modular multi-node ROS2 perception stack separating inference, depth estimation, and sensor fusion, enabling parallel subsystem development and clean integration with downstream flight controls

ROS Executive | University of Toronto Autonomous Rover Team

July 2023 - August 2024

- Integrated Cartographer SLAM and costmap-based path planning from the ROS Navigation Stack for the Intelligent Ground Vehicle Competition; fused LiDAR and camera data to generate accurate environmental maps for autonomous traversal
- Calibrated and filtered LiDAR and camera sensors, processing raw data into ROS messages and reducing odometry drift by 5%
- Developed ROSpy control scripts and tuned a PID path-tracking controller, reducing off-course veering by 20%; validated navigation behaviors end-to-end in Gazebo with a differential-drive URDF kinematic model

Embedded Lead | FlameBot, A Fire Surveyor and Mitigation Tool - Personal Project

February 2024 - April 2024

- Developed firmware using Embedded C in a real-time environment to interact with peripherals and utilized ADC HAL for optimal sensor data acquisition; used JTAG debugger to validate code efficacy
- Applied a PID control algorithm for land surveying (line tracking) and optimized firmware via fire-detection interrupt handling; actuated a servo pump for precise water ejection, yielding a 95% success rate